

PHOTOSYNTHESIS

Photosynthesis is the process by which green plants convert sunlight into chemical energy. It occurs mainly in the chloroplasts and uses chlorophyll to absorb light energy.

Inputs:

- Carbon dioxide (CO_2)
- Water (H_2O)
- Sunlight

Outputs:

- Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
- Oxygen (O_2)

Photosynthesis occurs in two stages:

- 1) Light-dependent reactions: capture sunlight and generate ATP and NADPH.
- 2) Calvin cycle (light-independent reactions): uses ATP and NADPH to produce glucose.

Importance:

Photosynthesis is essential for life on Earth because it produces oxygen and forms the base of food chains.